# runDIC quick start

CellScale biaxial test → stress–stretch data, in MATLAB. The full guide is `README.md` in the `matlab` folder.

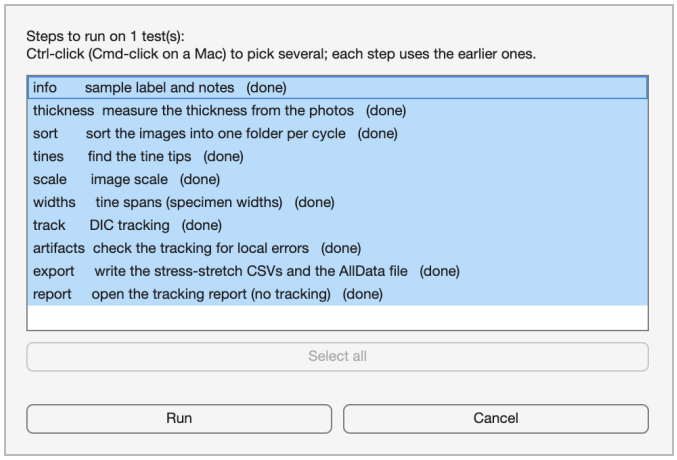
## Before the first run

- MATLAB R2025b or later; no toolboxes needed.
- Copy the `matlab` folder to your computer (for example `Documents\runDIC`) and make it MATLAB's current folder (the folder bar at the top).
- The data: a sample folder (e.g. `N1900_LFPA`) holding one folder per test, named `test1_…`, `test2_…`, with the tester's files as it wrote them, and the photos of the strips in `Thickness, <ID>\test <N>` (e.g. `Thickness, 1900\test 1`).

## Start

Type `runDIC` in the Command Window and choose the **test folder** (or the sample folder, to run all its tests). The step list opens: keep the checked steps and click **Run**. Then answer each window as below.

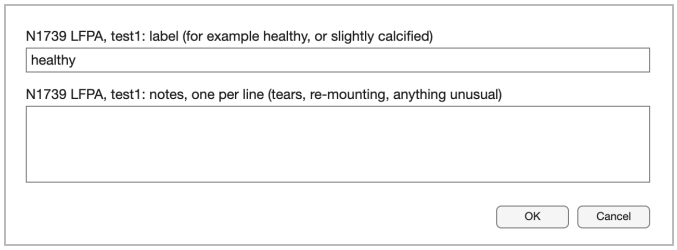
Every window has **Cancel**, which stops the whole run (closing a window does the same). Where a step's results already exist, its window also has **Skip**, which keeps them.




The step list. "(done)" marks what earlier runs did.

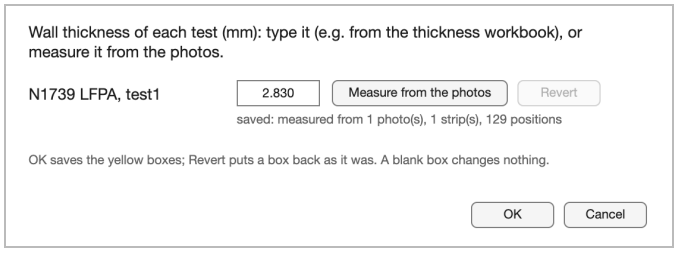
### 1 Label and notes

Type the sample's label (for example *healthy*) and note anything unusual: a tear, re-mounting, a bubble. Click OK.



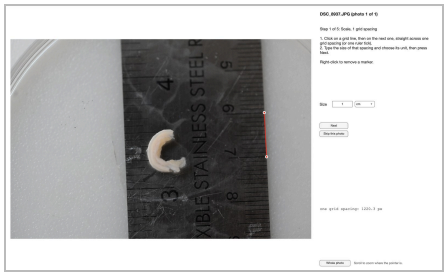

### 2 Thickness

One row per test. Type the thickness in mm (for example from the thickness workbook), or click **Measure from the photos** (below). A new value turns the box yellow; **Revert** puts it back. Click **OK** to save the yellow boxes. A test without a thickness gets empty stress columns.

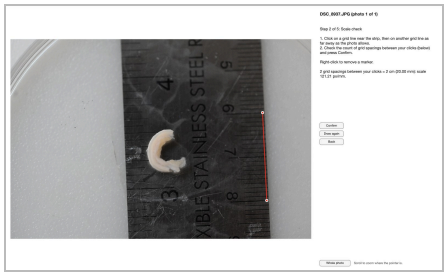


## Measuring the thickness from the photos

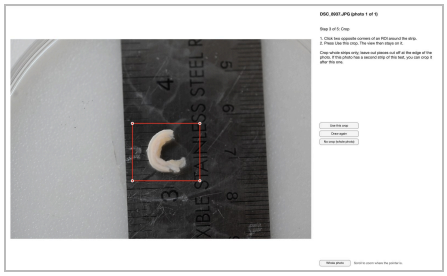
Each photo opens in turn. Your clicks are red; right-click removes the last one, and scrolling zooms where the pointer is. A double-click on the last point of a step does the same as its first button.



**Step 1: Scale, 1 grid spacing.** Click on a grid line, then on the next one, straight across one grid spacing. Type its size and unit (here 1 cm on a ruler; a mat square is **0.5 in**) and press **Next**.



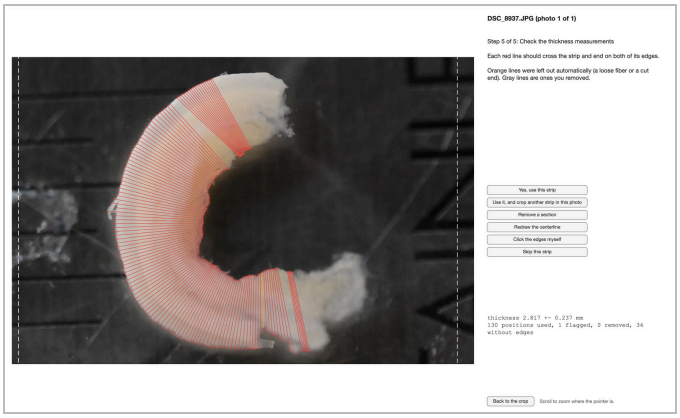
**Step 2: Scale check.** Click on a grid line near the strip, then on another one as far away as the photo allows. Check the count of grid spacings (it must be a whole number) and press **Confirm**; if a click missed a grid line, the window says so: **Draw again**.



**Step 3: Crop.** Click two opposite corners of an ROI around one whole strip, then **Use this crop**. Leave out pieces cut off by the photo's edge.



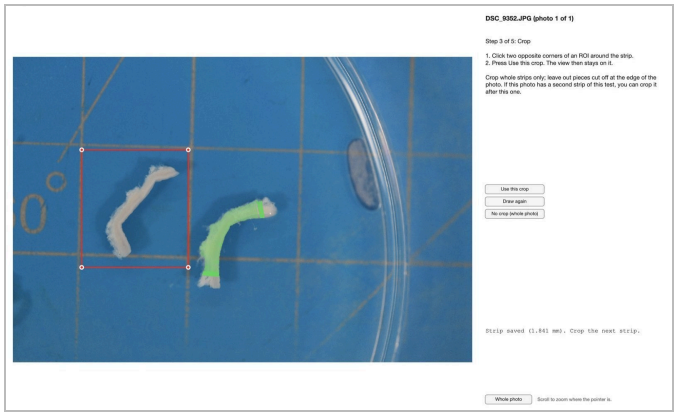
**Step 4: Trace the centerline.** Click along the middle of the strip, from one end to the other, then **Done**.



**Step 5: Check the thickness measurements.**

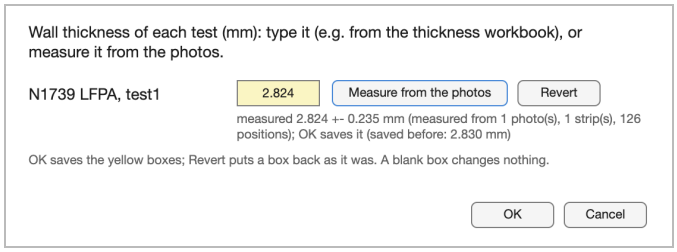
> In step 5, each red line should cross the strip and end on both of its edges. If they do, click **Yes, use this strip** (the next photo opens). Orange lines are left out automatically.

If not: **Remove a section** (click its two ends) for a bad stretch, **Redraw the centerline**, or **Click the edges myself** (along one edge, then the other). **Skip this strip** if nothing works. A strip needs at least 10 positions.




If the photo has a second strip of the test, **Use it, and crop another strip in this photo** keeps the first strip (green) and goes back to step 3 with the same scale.

When the last photo is done, the thickness window comes back with the result in yellow and the value saved before, if any. Click **OK** to save it, or **Revert** to keep the saved value.




## Checks for each test

### 3 Sorting check

Every cycle (Def01–Def19) with its images and the frames missing from the tester's files. With no red rows, click **Proceed**. A few missing frames (a red row with 3–5 missing) are fine: the camera drops frames now and then, so also **Proceed**. A folder with no images, or more missing, means something went wrong: click **Cancel** and ask your supervisor.

| Folder | Images | First image (s) | Last image (s) | Images missing | Data.csv rows missing | Coords1.csv rows missing | Note |
|---|---|---|---|---|---|---|---|
| Def01_Preconditioning_c01 | 380 | 0.0 | 75.8 | 0 | 0 | 0 | |
| Def02_Preconditioning_c02 | 380 | 76.0 | 151.8 | 0 | 0 | 0 | |
| Def03_Preconditioning_c03 | 380 | 152.0 | 227.8 | 0 | 0 | 0 | |
| Def04_Preconditioning_c04 | 380 | 228.0 | 303.8 | 0 | 0 | 0 | |
| Def05_Preconditioning_c05 | 380 | 304.0 | 379.8 | 0 | 0 | 0 | |
| Def06_Preconditioning_c06 | 380 | 380.0 | 455.8 | 0 | 0 | 0 | |
| Def07_Preconditioning_c07 | 380 | 456.0 | 531.8 | 0 | 0 | 0 | |
| Def08_Preconditioning_c08 | 380 | 532.0 | 607.8 | 0 | 0 | 0 | |
| Def09_Preconditioning_c09 | 380 | 608.0 | 683.8 | 0 | 0 | 0 | |
| Def10_Preconditioning_c10 | 380 | 684.0 | 759.8 | 0 | 0 | 0 | |
| Def11_Equibiaxial | 380 | 760.0 | 835.8 | 0 | 0 | 0 | |
| Def12_1-0.75_ratio | 286 | 836.0 | 892.9 | 0 | 0 | 0 | |
| Def13_1-0.5_ratio | 200 | 893.0 | 932.8 | 0 | 0 | 0 | |
| Def14_1-0.25_ratio | 200 | 933.0 | 972.8 | 0 | 0 | 0 | |
| Def15_Equibiaxial | 374 | 973.0 | 1048.8 | 6 | 0 | 6 | 6 image(s) missing: 1032.8-1033.8 s (all unloading); 6 Coords1.csv row |
| Def16_0.75-1_ratio | 380 | 1049.0 | 1124.8 | 0 | 0 | 0 | |
| Def17_0.5-1_ratio | 379 | 1125.0 | 1200.8 | 1 | 0 | 1 | 1 image(s) missing: 1174.6 s (all unloading); 1 Coords1.csv row(s) miss |
| Def18_0.25-1_ratio | 380 | 1201.0 | 1276.8 | 0 | 0 | 0 | |
| Def19_Equibiaxial | 381 | 1277.0 | 1353.0 | 0 | 0 | 0 | |

test1 08252026 PLC25 38-20: 6760 images, one every 0.20 s; 0 images outside every cycle (before the first or after the last).
Def15_Equibiaxial: 6 image(s) missing: 1032.8-1033.8 s (all unloading); 6 Coords1.csv row(s) missing (the same frames)
Def17_0.5-1_ratio: 1 image(s) missing: 1174.6 s (all unloading); 1 Coords1.csv row(s) missing (the same frames)

**Some folders are missing frames (red rows). A few missing frames are fine (the camera drops frames now and then), and frames missing during unloading don't affect the loading curves; a folder with no images or many missing means the test or the folders need a look. Proceed with this test?**

Proceed Cancel

The example test: Def15 and Def17 are red, with 6 and 1 frames missing during unloading. The loading curves are complete, so **Proceed**.

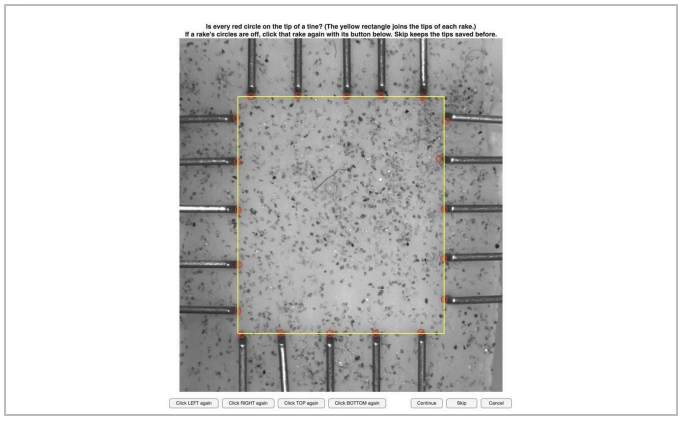
## 4 Tine tips

Every red circle should sit on the tip of a tine: the end of the dark rod, where it meets the tissue. Look closely: in this example, the second circle of the right rake sits out in the tissue, left of its tine. If every circle is right, **Continue**.

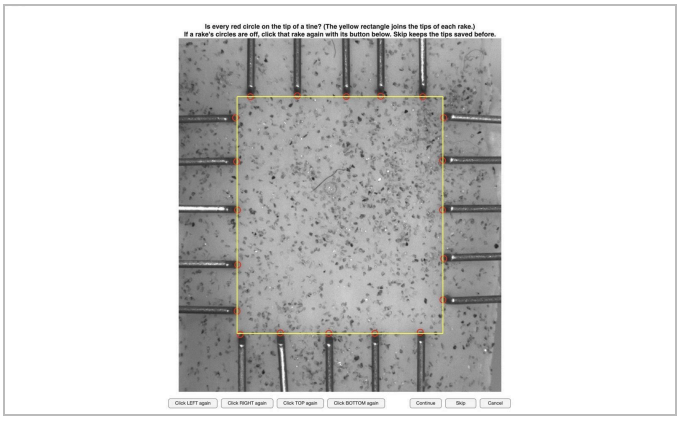
If a rake's circles are off, press its button (**Click LEFT again**, …) and click its 5 tips: the end of each rod, where it meets the tissue. The window opens zoomed on the rake; scroll to zoom further, right-click removes the last click, and **Done** takes the 5 clicks. The tips window then comes back with the new tips.

If the tips of a rake can't be found at all, the click window opens right away.

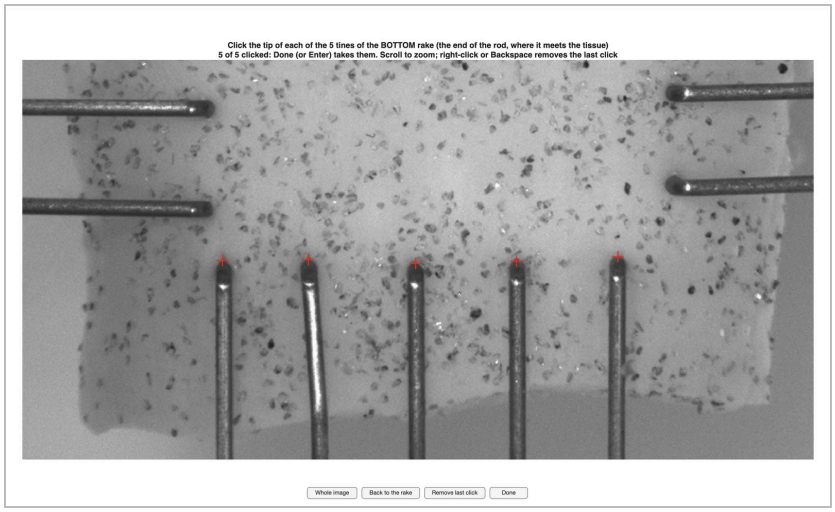
The image scale and the tine spans are then measured by themselves.




As found: the second circle of the right rake is off.

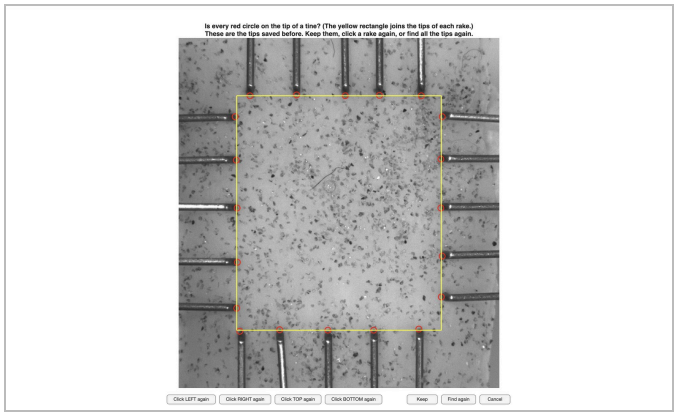



After **Click RIGHT again**: every circle on a tip.




Clicking the tips of one rake: each red cross at the end of a rod.

When the test already has tine tips from an earlier run, the window shows those instead of searching again. **Keep** keeps them; a rake's button clicks that rake again; **Find again** searches the image as for a new test.

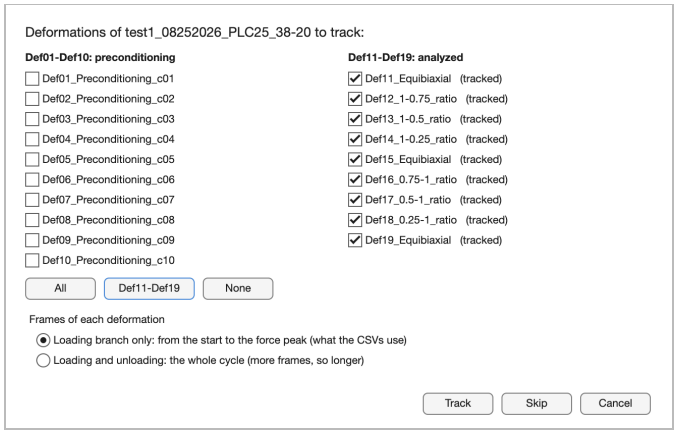


# Tracking and export

## 5 What to track

Keep the checked deformations (Def11–Def19) and **Loading branch only**, and click **Track**. "(tracked)" marks the ones tracked before. **All** adds the preconditioning cycles, Def01–Def10: only for checking (the export can then plot them on their own), and the tracking takes about twice as long.

The run then proceeds by itself, about 15 minutes per test (longer on a laptop). The Command Window shows each step and, during tracking, each deformation's progress:



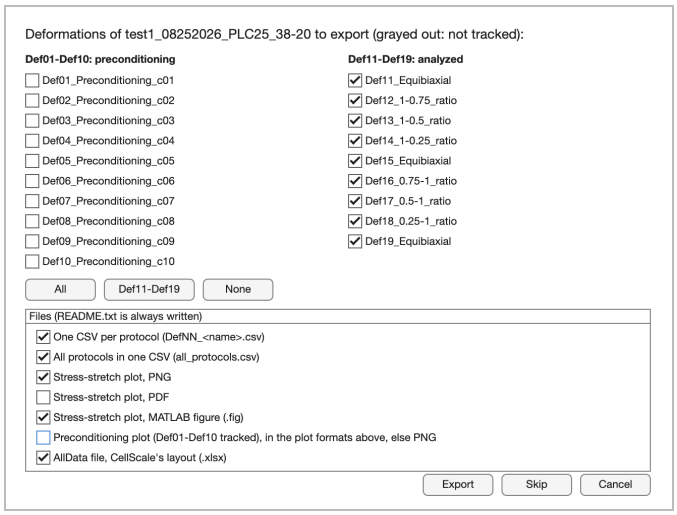

```
-- image scale
  Def01_Preconditioning_c01: tracking 20 tine tips in 380 frames
  Def19_Equibiaxial: tracking 20 tine tips in 381 frames
  scale: 62.882 px/mm (X 62.939, Y 62.824; SD across pairs 2.08%); left out:
-- tine spans
  tine spans: left 9.93, right 9.20, top 8.80, bottom 9.16 mm -> w_theta 9.57, w_z 8.98 mm
-- tracking Def01–Def19, loading branches only
  reference Def11_Equibiaxial t = 760.0 s; grid 41 x 47 points; subset 31 px (1.03 mm), step 6 px (0.20 mm); images x0.477 to 30.0 px/mm; saturated glints masked
  [1/19] Def01_Preconditioning_c01: 191 frames (loading)
    Def01  191/191 frames  100%  0m22s elapsed, 0m00s left  (valid 99.1%, ZNCC 0.9960)
    Def01_Preconditioning_c01 done: valid per frame median 100.0%, min 99.0%; bridge valid 100.0%
  [2/19] Def02_Preconditioning_c02: 191 frames (loading)
    Def02  191/191 frames  100%  0m21s elapsed, 0m00s left  (valid 99.1%, ZNCC 0.9959)
    Def02_Preconditioning_c02 done: valid per frame median 100.0%, min 99.1%; bridge valid 100.0%
  [3/19] Def03_Preconditioning_c03: 191 frames (loading)
    Def03  168/191 frames   88%  0m19s elapsed, 0m03s left  (valid 99.9%, ZNCC 0.9950)
```

The Command Window of the example test: the image scale, the tine spans, then the tracking, one deformation at a time (here with **All**, Def01–Def19). Valid points stay near 100%.

## 6 What to export

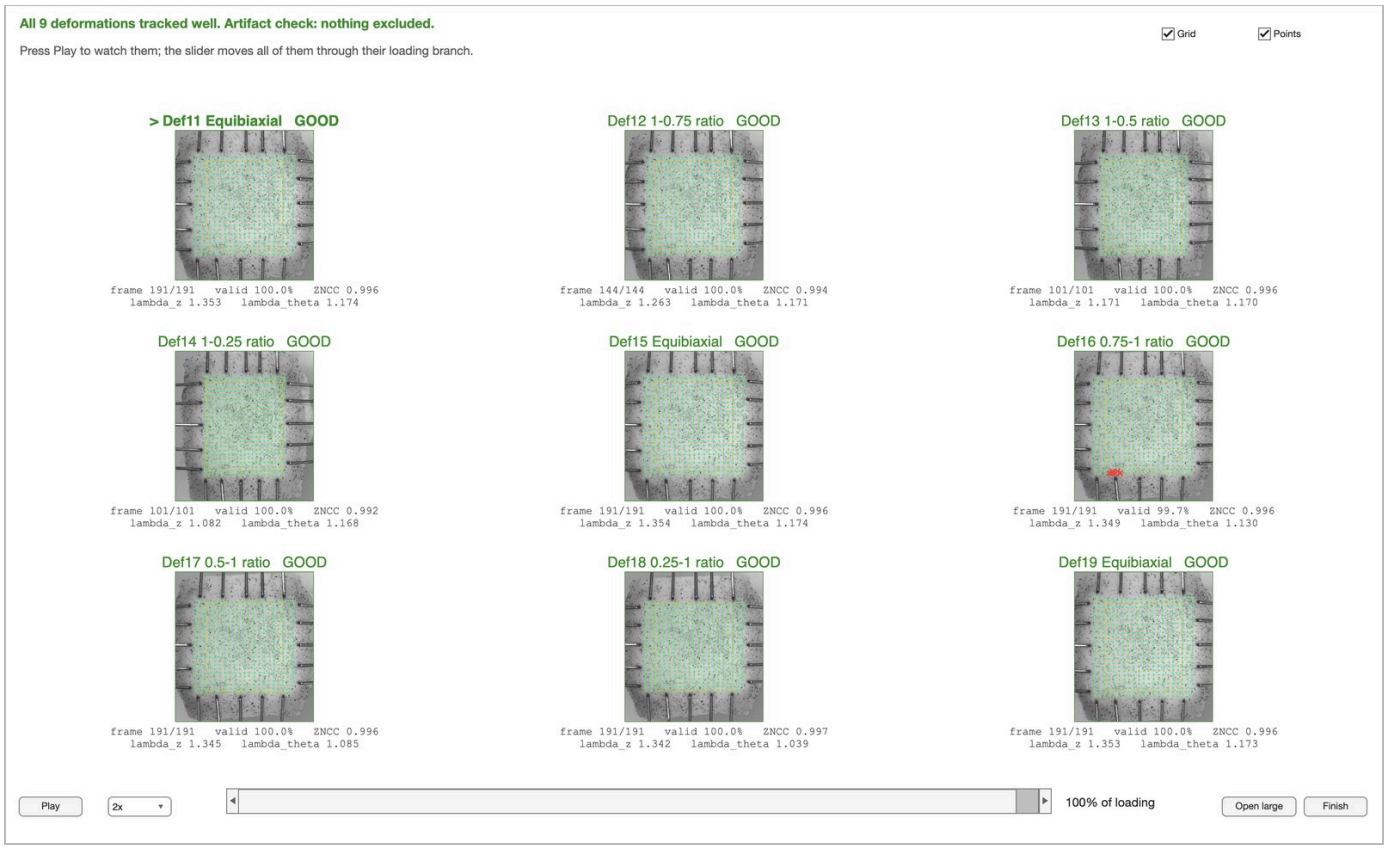
When the tracking is done, pick the deformations (Def11–Def19 by default; grayed out: not tracked) and the files to write, and click **Export**. The run waits here for your answer.

When the preconditioning cycles were tracked, **Preconditioning plot** draws them on a plot of their own; otherwise it is grayed out.

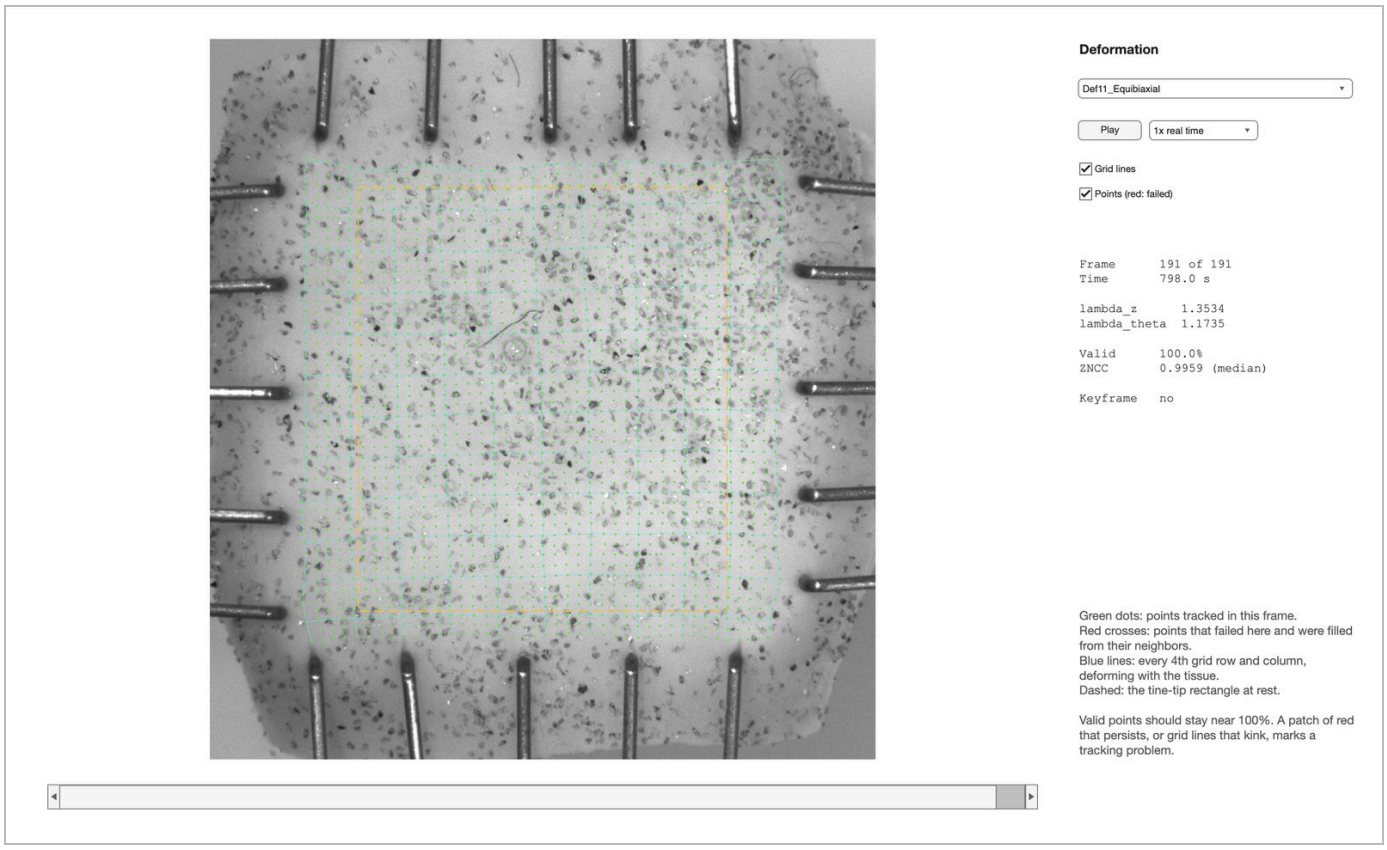



## 7 Tracking report

At the end, the report shows Def11–Def19 side by side; **Play** runs them. Green dots are tracked points, red crosses failed ones. Each tile is marked GOOD, CHECK or PROBLEM; **Open large** shows the selected one big, frame by frame, and **Finish** closes the report.




The report of the example test: all 9 GOOD, a few scattered red crosses.

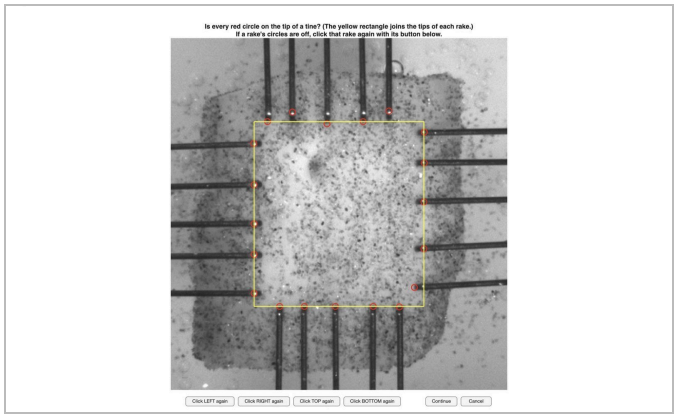



**Open large**: one deformation, frame by frame, with its stretches and the share of valid points.

## Recognizing problems

The example test above went well. The pictures below are from another test, with three typical problems.

**Tine tips off** (step 4). Circles on a shiny spot of the rod or out in the tissue: here two circles of the top rake sit on the shiny spot partway up the rod, and the lowest of the right rake is out in the tissue. Click that rake again; the image scale and the tine spans are measured from these tips, so wrong tips make the stresses wrong.




**Tine pairs left out of the scale** (Command Window, `-- image scale`). One or two `left out tine pair` lines are normal. Several, or `X NaN` / `Y NaN` in the `scale:` line (no pair left on that axis), mean the scale is unreliable: check the tine tips (run the **tines** step again) and tell your supervisor. Here four pairs are left out and the X scale is missing; the tracking also keeps only 69% of the points valid.

```
-- image scale
  Def01_Preconditioning_c01: tracking 20 tine tips in 400 frames
  Def19_Equibiaxial: tracking 20 tine tips in 401 frames
  left out tine pair X2 (28.35 px/mm, -19.4% from the median 35.18)
  left out tine pair X3 (31.52 px/mm, -10.4% from the median 35.18)
  left out tine pair X4 (29.98 px/mm, -14.8% from the median 35.18)
  left out tine pair Y5 (29.58 px/mm, -15.9% from the median 35.18)
  scale: 35.127 px/mm (X NaN, Y 35.127; SD across pairs 1.79%); left out: X2, X3, X4, Y5
-- tine spans
  tine spans: left 9.51, right 9.85, top 7.69, bottom 7.65 mm -> w_theta 9.68, w_z 7.67 mm
-- tracking Def11-Def19, loading branches only
  reference Def11_Equibiaxial t = 800.0 s; grid 39 x 47 points; subset 35 px (1.00 mm), step 7 px (0.20 m
  [1/9] Def11_Equibiaxial: 201 frames (loading)
    Def11  201/201 frames  100%  0m28s elapsed, 0m00s left  (valid 69.3%, ZNCC 0.9871)
    Def11_Equibiaxial done: valid per frame median 87.9%, min 69.3%; bridge valid 99.7%
```

**A red patch in the tracking report** (step 7). Red crosses that come and go are normal. A large patch that persists, as in all 9 tiles here, means the tracking failed there, usually where the speckle is too sparse or the tissue is bare. Those deformations are marked PROBLEM or CHECK; don't use their results without your supervisor.




> **Stop and ask your supervisor** before using the results when:
>
> - the sorting table has a folder with no images, or more than 5 frames missing;
> - tine tips can't be put right by clicking, or the scale has several tine pairs left out (or X NaN / Y NaN);
> - the thickness edges stay off the strip;
> - the Command Window says an artifact was excluded, or the report has any CHECK or PROBLEM tile.

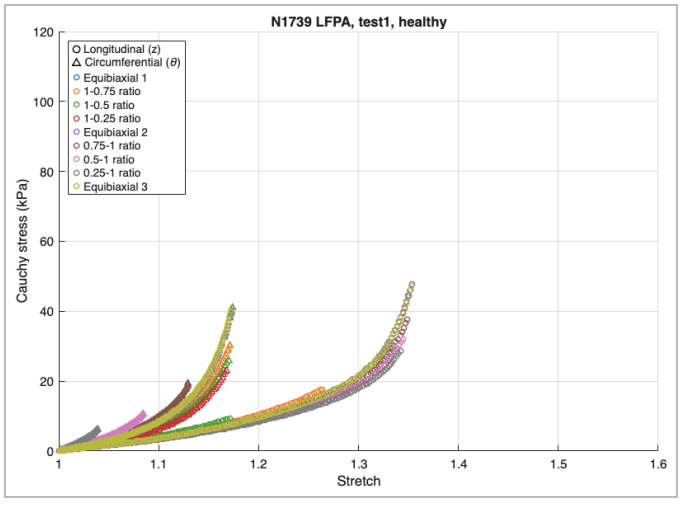
## If something goes wrong

| | |
|---|---|
| "… excluded … tracking again" in the Command Window | a local tracking error was found and left out; the test is tracked a second time, so the run takes about twice as long. Expected; tell your supervisor. |
| `=== test… FAILED: <message>` | that test stopped; write down the message and ask your supervisor. With a sample folder, the other tests still run. |
| Stopping a run | **Cancel** in any window, or **Ctrl-C** in the Command Window during tracking. Run `runDIC` again: the step list marks the steps already done; select the rest. A tracking step that was interrupted starts over. |

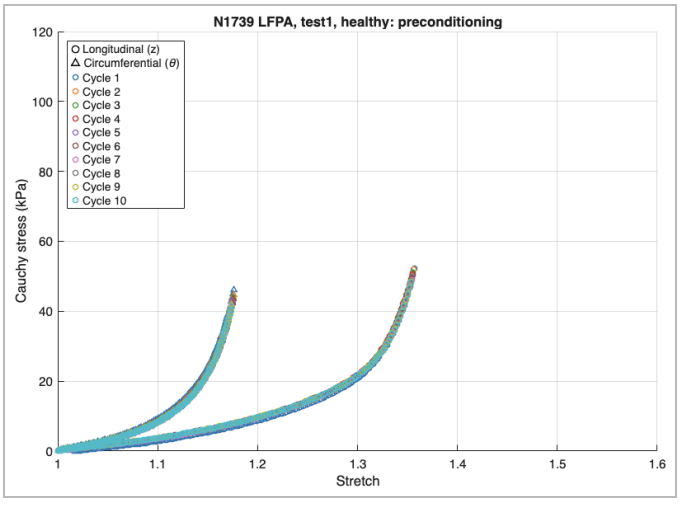
## The results

In the test's `stress_stretch` folder, the files picked in the export window:

| | |
|---|---|
| `stress-stretch test<N>.AllData.xlsx` | for your modeling script: the layout of CellScale's AllData export, with the DIC stretches and stresses |
| `stress_stretch.png`, `.pdf`, `.fig` | every protocol on one plot, to check at a glance (the `.fig` opens in MATLAB) |
| `stress_stretch_preconditioning.png`, … | the preconditioning cycles on their own plot, when picked |
| `all_protocols.csv`, `DefNN_….csv` | the same data as CSV files |
| `README.txt` | what was used: protocols, thickness, widths, scale, tracking verdict |




`stress_stretch.png` of the example test: circles longitudinal, triangles circumferential, one color per protocol.




The preconditioning plot of the same test: Cycle 1 to Cycle 10.

## Common tasks

| | |
|---|---|
| Open the tracking report again | type `viewTracking` and choose the test folder |
| Change the thickness, without tracking again | `runDIC`, choose the folder, select only **thickness** and **export** |
| Export other protocols or files | `runDIC`, choose the folder, select only **export** |
| Redo a step | `runDIC`, select that step and the ones after it |